Independent replication report:

On-lattice kinetic Monte Carlo approaches for modeling molecular anisotropy in resveratrol crystallization

Janicki et al., *Modelling Simul. Mater. Sci. Eng.* **33** 055010 (2025)

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X-100 Replication Project (agent, OSTI wave, 2026-07-06)

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Abstract

We independently reproduce as much of Janicki et al. (2025) as the public artifacts support. The two SPPARKS branches referenced by the paper (`tdjanic-snl/spparks`, branches `nonorth` and `resveratrol`) are live on GitHub. The `resveratrol` branch compiles cleanly on a Linux workstation with system mpicxx and `-std=c++17`. The infrastructure claims (HCP lattice, hex region, 3D random deposition mode) build and run at paper-scale (48×16×24 HCP unit cells, ≈36,864 sites) with the paper's parameters ($T = 0.0270$ eV/k_B, $\nu = 0.1$ s^{-1}, capture 5.0 Å, coord [1,9], tree solver). However, the paper's headline software addition — the `diffusion/disphere` app style and the `resv` lattice style described in Section 2.2 — *are not actually present* in the public `resveratrol` branch as of commit `f6bcc3b` (2025-03-31). The 74-point DFT binding-energy library and the input scripts promised in "Supplemental Materials" are behind a captcha-blocked IOP link. A 10-seed control sweep with an isotropic monotonic Arrhenius ladder produces distinctly more isotropic aspect ratios ($W{:}L = 0.53 \pm 0.03$, $H{:}L = 0.87 \pm 0.04$) than the paper's experimental peaks ($W{:}L \approx 0.35$, $H{:}L \approx 0.55$) — consistent with the paper's central claim that anisotropy requires the bound-sphere logic, but not itself a positive test of Figure 7. Verdict: **PARTIAL**.

Contents

1 Paper summary

1.1 Method as described

The paper introduces two enhancements to the Sandia SPPARKS on-lattice KMC package:

1. **Non-orthogonal (HCP / hex) simulation boxes.** A new `lattice HCP a` style, a new `region hex ...` region style, and a new 3D-random deposition mode of `deposition event ...` in which passing incident vector $(0, 0, 0)$ replaces the default top-face restriction with random sampling of candidate deposition points anywhere in the box.
2. **Bound-sphere disphere app.** A new `diffusion/disphere` app style and a new `lattice resv` lattice style in which each resveratrol molecule is represented by two hard spheres bound to a nearest-neighbor pair. Coordination-indexed binding energies (`ecoord`) become 2D-indexed to account for the two halves of the monomer with different OH substituents.

The rate expression is the standard Arrhenius form,

$$\Gamma(i) = \nu \exp\left[\frac{E_{\text{bind}}(i)}{k_{\text{B}}T}\right], \tag{1}$$

with $E_{\text{bind}}(i)$ tabulated by coordination and ν chosen from experimental precursor flow/concentration.

The enhancements are applied to *trans*-resveratrol ($P2_1/c$) with a 74-entry DFT binding-energy library (FHI-aims, PBE + Tkatchenko-Scheffler vdW, tier-1 basis, "light" settings, 3×3×1 k-mesh, >20 Å vacuum, bottom layer(s) fixed). Production simulations use a 48×16×24-unit-cell HCP box ($\approx 200 \times 250 \times 210$ nm) at $T = 20\,°\text{C}$ ($k_{\text{B}}T = 0.0270$ eV), $\nu = 0.1\ \text{s}^{-1}$, capture 5.0 Å, coord $[1, 9]$, `tree` solver, 5×10^6 trial steps. 5 initial-nucleus geometries × 112 seeds = 560 replicates.

1.2 Experimental validation as described

2 g resveratrol dissolved in 20 g 2:1 ethanol/water at 70 °C for 24 h, cooled to 20 °C for 5 days. Micro-CT (Bruker SKYSCAN 1272, 0.45 μm) with Dragonfly 2024.1 watershed segmentation into 1207 crystals. Aspect-ratio comparison of $H:L$ and $W:L$ histograms (Figure 7). Paper reports "peaks overlap well".

2 Claims table

ID	Claim	Type	Testable?	Tested here?	
C1	`nonorth` branch adds working HCP lattice, hex region, and 3D-random deposition.	Software	Yes	Yes REPLICATED	—

ID	Claim	Type	Testable?	Tested here?
C2	HCP + hex-region + 3D-random-deposition run stably on the paper's $48 \times 16 \times 24$ paper-scale box with the paper's parameters.	Computational	Yes	Yes — REPLICATED (10 seeds, 0 crashes)
C3	`resveratrol` branch adds `diffusion/disphere` app style and `resv` lattice style.	Software	Yes (by inspection)	**Yes — CONTRADICTED on the current public branch state**
C4	74-entry DFT (FHI-aims / PBE + TS-vdW) binding-energy library parametrizes the KMC model.	Comp/Data	In principle	**No — data blocked** (embargo on public code, captcha-blocked supplementary)
C5	KMC with bound-sphere + DFT ecoord produces aspect-ratio distributions whose peaks overlap experimental peaks (Figure 7).	Empirical	Requires C3+C4	**No** — only a consistency spot-check with an isotropic surrogate
C6	Nucleus-shape sensitivity (5 nucleus geometries in Table 3) broadens the simulated H:L distribution.	Empirical / methodological	With C3+C4	**No**

3 Method used in this replication

1. **Fetch.** `curl https://www.osti.gov/servlets/purl/2583708 -o paper.pdf` on uicgpu (1,304,946 B). scp back.
2. **Extract.** `pdftotext -layout` (poppler 26.06.0) → `work/paper_layout.txt`; publisher HTML scrape for table structure. Marker/Nougat not available on uicgpu; produced Marker-flavored (`extraction/marker.md`) and Nougat-flavored (`extraction/nougat.mmd`) extractions from the pdftotext+HTML output, with a stated fidelity caveat.
3. **Discover public code.** GitHub API → branches `master`, `nonorth`, `resveratrol`. Cloned all three.
4. **Diff.** `git diff master resveratrol -- src/` shows 31 files changed, +577/−133 lines. Additions: `lattice.cpp` (HCP style only), `region_hex.{cpp,h}` (new), `create_sites.cpp` (HCP + 3D deposition), `app_diffusion.cpp` (3D random deposition mode). *No* disphere app. *No* resv lattice.

5. **Build.** Copied `spparks-resv/` to uicgpu, wrote `src/MAKE/Makefile.uic` (mpicxx wrapper, `-std=c++17 -O2`, no libjpeg). `make uic -j 8`: clean; produced `spk_uic` (895 KB). SPPARKS reports version banner “27 Nov 2024”.

6. **Smoke.** `app_style diffusion nonlinear hop` + `lattice hcp 1.0` + `region mybox hex 0 8 0 8 0 4` + 3D-random deposition. Passes.

7. **Paper-scale sweep.** 10 seeds, box `hex 0 48 0 16 0 24` (36,864 sites, box $56\times 13.86\times 39.19$ with xy-tilt 8, matching primitive HCP for a $48\times 16\times 24$ cell replication). Nucleus `block 20 28 6 10 10 14`. $T = 0.0270$, $\nu = 0.1$, capture 5.0, coord [1,9], `tree` solver, 2000 KMC time-units. Isotropic monotonic ecoord ladder $n \to -0.2n$ eV for $n \in [1, 12]$ (surrogate, because the paper's 74-entry DFT library is not accessible).

8. **Analyze.** Python parses the final-frame dump, extracts positions of `OCCUPIED` (`i1==2`) sites, computes axis spans, sorts to give L (largest), M (middle), S (smallest), then $W\!:\!L = S/L$, $H\!:\!L = M/L$.

4 Results vs paper

4.1 Infrastructure claims C1, C2

- **Compile.** Clean, 0 errors, 0 warnings. Produces `spk_uic` that identifies itself as “SPPARKS 27 Nov 2024”.
- **HCP + hex region.** 512-site smoke box has expected $2\times 8\times 8\times 4$ site count and 12 neighbors per site. Box dimensions match $\sqrt{3}/2 \cdot N_y$ and $\sqrt{8/3} \cdot N_z$ for the ideal-HCP primitive cell.
- **3D-random deposition.** Incident vector $(0, 0, 0)$ is parsed and runs; deposition acceptance grows linearly in time for our surrogate ladder.

4.2 Aspect-ratio measurement (surrogate ladder, 10 seeds)

Seed	n_{occ}	span$_x$	span$_y$	span$_z$	L	M	S	$W\!:\!L$	$H\!:\!L$
1	390	10.0	5.196	8.165	10.0	8.165	5.196	0.520	0.816
2	380	10.0	4.907	8.165	10.0	8.165	4.907	0.491	0.816
3	377	9.5	5.196	8.165	9.5	8.165	5.196	0.547	0.859
4	385	10.0	5.196	8.982	10.0	8.982	5.196	0.520	0.898
5	380	9.0	4.619	8.165	9.0	8.165	4.619	0.513	0.907
6	380	10.0	4.907	8.165	10.0	8.165	4.907	0.491	0.816
7	381	9.0	5.196	8.165	9.0	8.165	5.196	0.577	0.907
8	384	9.0	4.907	8.165	9.0	8.165	4.907	0.545	0.907
9	384	9.5	5.196	8.165	9.5	8.165	5.196	0.547	0.859
10	373	9.0	5.196	8.165	9.0	8.165	5.196	0.577	0.907
mean	381							**0.533**	**0.870**
std	±5							**±0.031**	**±0.041**

Paper Figure 7 (visual read of PDF): experimental peaks near $W\!:\!L \approx 0.30$–0.40 and $H\!:\!L \approx 0.50$–0.60. Our isotropic-surrogate is at $W\!:\!L = 0.53$, $H\!:\!L = 0.87$ — both distinctly more isotropic (higher $W:L$, higher $H:L$) than the experimental peaks. This is exactly the behaviour the paper predicts should happen without the bound-sphere anisotropy logic. So the result is *consistent with* the paper's central claim, but is not a positive test of Figure 7.

4.3 Contradiction on C3

The paper's Section 2.2 explicitly claims a `diffusion/disphere` app style and a `resv` lattice style are present in the `resveratrol` branch. Neither string appears in the source tree of the public `resveratrol` branch:

Listing 1: grep of the public resveratrol branch source

```
$ grep -riIn "disphere\|resv\|bound.sphere" src/*.cpp src/*.h
(no matches)
$ grep -n "strcmp(arg\[0\]" src/lattice.cpp
39:  if (strcmp(arg[0],"none") == 0) style = NONE;
40:  else if (strcmp(arg[0],"line/2n") == 0) style = LINE_2N;
41:  else if (strcmp(arg[0],"sq/4n") == 0) style = SQ_4N;
42:  else if (strcmp(arg[0],"sq/8n") == 0) style = SQ_8N;
43:  else if (strcmp(arg[0],"tri") == 0) style = TRI;
44:  else if (strcmp(arg[0],"hcp") == 0) style = HCP;
45:  else if (strcmp(arg[0],"sc/6n") == 0) style = SC_6N;
    ...  (no "resv") ...
```

Best-guess explanation: the paper's Data Availability Statement says "*will be* openly available following an embargo" — so this is an in-progress embargo release rather than a claim that was falsified. The infrastructure was pushed; the scientific extensions were not (yet).

5 Verdict

PARTIAL.

- C1, C2 (infrastructure): **REPLICATED**.
- C3 (disphere/resv software claim): **CONTRADICTED** on the current public artifact state.
- C4–C6 (DFT ecoord, aspect-ratio validation, nucleus-shape sensitivity): **BLOCKED** by missing supplementary and missing disphere code. Only a consistency spot-check is possible with the isotropic surrogate, and that spot-check is consistent with (but not a positive test of) the paper.

We replicate half the paper's method claim and none of the specific numerical Figure 7 comparison.

6 Open Questions

Q1. Ablation: how much of the reported anisotropy in Figure 7 is the bound-sphere geometric constraint per se, and how much is the 2D-indexed DFT ecoord table? Section 2.2 conflates two independent mechanisms and no ablation is reported.

Q2. Sensitivity: does the "arbitrarily large barrier" imputation for un-tabulated coordinations bias the reported facet ordering toward the specific 74-point sampling geometry of the DFT library? No sensitivity study is reported.

Q3. Unit-cell ↔ nm mapping: from $P2_1/c$ resveratrol cell parameters (a≈7.6 Å, b≈24.5 Å, c≈15.9 Å), a 48×16×24 replication is $\approx 365 \times 392 \times 382$ nm, 1.5–2× the stated 200×250×210 nm. Which is correct?

Q4. Convergence: is $5{\times}10^6$ trial steps per run $\times$ 560 replicates enough to converge the aspect-ratio histogram, or does the reported "much narrower simulated distribution than experiment" reflect under-sampling?

Q5. Inverse problem: can an isotropic 1D ecoord ladder inverse-fit to Micro-CT aspect ratios approximately recover the DFT 2D rate table, and if so how much of the parametric machinery is compressible?

Each question with concrete next-steps is in `report/open_questions.json`.

7 Data and code

- SPPARKS `resveratrol` branch: https://github.com/tdjanic-snl/spparks/tree/resveratrol
- SPPARKS `nonorth` branch: https://github.com/tdjanic-snl/spparks/tree/nonorth
- Upstream SPPARKS: https://github.com/spparks/spparks
- OSTI PDF: https://www.osti.gov/servlets/purl/2583708
- IOP publisher page: https://iopscience.iop.org/article/10.1088/1361-651X/ade176

Local artifacts in this replication target directory (see `report/artifacts_summary.md`).